

Notes: Giambona, Kumar, and Phillips (RFS, 2025)

Hedging, Contract Enforceability, and Competition

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1 Big picture

- **Question.** The paper asks how lowering the contracting cost of hedging changes firm risk management, financial stability, product-market outcomes, and competition.
- **Setting.** The empirical setting is the U.S. life insurance industry over 2000–2017. This industry is large, systemically important, and especially relevant because life insurers sell long-duration products whose value to customers depends on the insurer remaining solvent.
- **Policy shock.** The key shock is the staggered state-level adoption of Section 711 of the Insurer Receivership Model Act. Section 711 grants derivatives counterparties superpriority rights in insurer insolvency, allowing them to terminate derivatives contracts and claim posted collateral immediately.
- **Economic channel.** Superpriority strengthens contract enforceability for derivatives counterparties. This lowers counterparty concerns when trading with riskier insurers, reduces the effective cost of hedging for financially constrained insurers, and expands their access to derivatives.
- **Main treatment group.** The most affected firms are insurers likely to face costly external finance before the reform. The paper measures this primarily with high pre-event leverage, and also uses low pre-event distance-to-default.
- **Main hedging result.** After Section 711, high-leverage insurers increase derivatives usage. In the preferred Table 3 specifications, derivatives notional rises by roughly $\exp(1.627) - 1 \approx 409\%$, the probability of using derivatives rises by 6.8 percentage points, and the hedging ratio rises by 25.3 percentage points.
- **Risk result.** Treated insurers become safer: the probability of large negative capital-and-surplus shocks falls, the probability of exit due to receivership or other events falls, income volatility declines, and ratings/financial-strength measures improve.
- **Product-market result.** Treated insurers sell more life insurance and annuities. They also lower prices for several life and annuity products and gain market share.
- **Competition result.** The reform changes product-market competition because risk management improves the competitive position of previously financially constrained insurers. The effect is stronger in more concentrated markets and translates into increased market-share outcomes.
- **Policy implication.** Derivatives superpriority can reduce systemic risk by making hedging more accessible to fragile financial firms, but it also changes competitive outcomes by helping financially constrained firms compete more aggressively.

2 Institutional background and mechanism

2.1 Why life insurance is the right laboratory

- Life insurers hold large asset portfolios and write long-duration liabilities. Their solvency matters not only to investors, but also to policyholders who rely on future payouts.
- Insurance customers may care about firm financial health because the contract is long lived. If an insurer is perceived as more likely to fail, customers may demand a discount or avoid buying its policies.
- The paper notes that derivative usage is concentrated in the life-insurance segment. Among insurance segments, life insurers account for the overwhelming majority of derivatives notional.
- Life insurers use derivatives primarily for hedging, not merely speculation. The paper reports that about 95% of life insurers' derivatives are used for hedging.

Interpretation: The industry combines three useful features: large derivative usage, meaningful solvency risk, and product demand that depends on perceived financial stability. This makes it ideal for asking whether hedging affects competition.

2.2 The reform: Section 711 of the Insurer Receivership Model Act

- Before the reform, a derivatives counterparty of a distressed insurer could face uncertainty, delay, or restrictions in insolvency proceedings.
- Section 711 provides a safe-harbor/superpriority treatment for derivatives contracts in insurer receivership.
- The reform allows nondefaulting derivatives counterparties to close out derivatives positions and claim collateral immediately if the insurer enters default or receivership.
- This makes derivatives contracts more enforceable and less risky for counterparties.
- Since the reform is adopted at different times by different states, the paper uses staggered adoption as a source of variation.

Interpretation: The reform does not directly subsidize insurers or mandate hedging. Instead, it changes the legal priority of derivatives contracts in insolvency. That makes it a clean shock to the enforceability and cost of hedging contracts.

2.3 Why contract enforceability changes hedging

- A derivatives counterparty worries about default exposure, collateral access, and legal delay.
- If counterparty protection is weak, counterparties charge more, demand more collateral, or refuse to trade with financially weaker insurers.
- Superpriority lowers counterparty expected loss and bargaining frictions.
- This is especially important for insurers that are ex ante closer to financial distress, because counterparties are most concerned about those firms.

Interpretation: The reform is predicted to matter mainly for constrained or risky insurers. Safe insurers already have easier access to derivatives, so the marginal benefit of stronger enforceability is smaller for them.

2.4 Conceptual product-market mechanism

The paper's product-market mechanism has three linked parts:

1. **Hedging channel.** Section 711 lowers the cost of derivatives trading for constrained insurers, so they hedge more.
2. **Financial-stability channel.** More hedging reduces exposure to financial shocks and lowers the likelihood of financial distress.
3. **Competition channel.** Safer insurers can lower prices, sell more policies, and gain market share because customers are more willing to buy from them and because their marginal cost of bearing risk is lower.

Interpretation: Figure 1 visualizes this idea. Demand shifts outward because customers value lower default risk. Supply shifts outward/downward because the insurer's financial input costs and risk-bearing costs fall. The new equilibrium has higher policy quantity and lower price.

2.5 The role of financial constraints

- The paper draws on the corporate hedging literature, especially theories in which financially constrained firms hedge less because hedging consumes scarce collateral.

- When legal reform raises the enforceability of derivatives, constrained firms become more able to hedge.
- The relevant firms are not already failed firms; they are firms that are solvent but more financially fragile than peers.
- This distinction is important because the product-market effect requires customers to respond to improved financial stability of operating insurers.

Interpretation: The paper is not merely about whether derivatives are used. It is about whether improving the contracting environment for derivatives changes real market behavior.

3 Data and measurement

3.1 Data sources

- The paper uses life-insurance company data from SNL and regulatory insurance filings for 2000–2017.
- Derivatives data come from NAIC Schedule DB, which reports derivatives positions and notional amounts.
- State-level adoption of Section 711 is collected from state legal changes to insurer receivership law.
- Financial-strength ratings are from A.M. Best; additional parent-company data come from S&P Global Companies.
- Product-price data are collected for life and annuity products to examine pricing effects, not only premium quantities.

Interpretation: The data allow the paper to connect a legal change to derivatives usage, insurer financial health, product sales, market share, and prices.

3.2 Treatment and timing variables

The main post-reform indicator is

$$PostSection711_{i,t} = 1 \tag{1}$$

if insurer i is domiciled in a state that has adopted Section 711 by year t ; it equals zero otherwise.

The main treatment interaction is

$$PreEventHighLeverage_i \times PostSection711_{i,t}. \tag{2}$$

- $PreEventHighLeverage_i$ equals one if the insurer’s leverage is above the sample median in the year before the insurer’s domicile state adopts Section 711.
- The alternative treatment proxy is $PreEventLowDistanceToDefault_i$, based on whether distance-to-default is below the sample median.
- Insurers in states that never adopt Section 711 and low-leverage insurers in adopting states help form the comparison group.

Interpretation: The design asks whether insurers most likely to benefit from a reduction in hedging costs respond more strongly after their domicile state adopts Section 711.

3.3 Hedging outcomes

The paper measures hedging using three main variables:

- **Log derivatives notional:** log of annual derivatives notional value.
- **Derivatives indicator:** whether the insurer reports any derivatives notional.
- **Hedging ratio:** derivatives notional divided by assets minus capital & surplus, interpreted as a normalized measure of hedging relative to regulatory liabilities.

Interpretation: Using all three variables distinguishes intensive-margin hedging, extensive-margin adoption of derivatives, and balance-sheet-scaled hedging exposure.

3.4 Financial-health outcomes

The paper studies multiple measures of stability:

- large negative shocks to capital & surplus;
- exit due to receivership or other events;
- hedging-adjusted income volatility;
- low A.M. Best rating;
- loss of top A.M. Best rating;
- low IRIS metric;
- low RBC ratio;
- parent-company put option price.

Interpretation: The paper's mechanism requires hedging to improve financial stability. The authors therefore use several financial-health proxies, not just derivative usage.

3.5 Product-market outcomes

The main product-market outcomes are:

- log life insurance premiums;
- log annuities;
- log number of policies;
- prices of 10-year term life, life annuity, and term annuity products;
- market share by state-year;
- top-market-share indicators;
- operating income and net income.

Interpretation: The paper moves from financial contracting to real outcomes. The key claim is that cheaper hedging affects competition because safer insurers sell more and price more aggressively.

3.6 Sample construction and summary statistics

- The sample period is 2000–2017 because Schedule DB derivatives data are available starting in 2000.
- The insurer-state premium sample has hundreds of thousands of observations because insurers operate across multiple states.
- Summary statistics show substantial skewness: median premiums and derivative notionals are much lower than means, reflecting a fragmented industry with a few very large insurers.
- At the insurer level, only about 20% of insurer-year observations report derivatives usage; among treated high-leverage insurers, the derivatives-use rate is higher, about 35%.
- The treated group is economically larger and more levered than the broader sample, consistent with the paper’s focus on insurers for which external finance is more costly.

Interpretation: The industry is heterogeneous. That heterogeneity is important for the identification strategy because the reform should affect financially fragile insurers more strongly than safer ones.

4 Empirical design and specifications

4.1 Baseline hedging regression

The paper estimates the following insurer-level regression:

$$\begin{aligned} Hedging_{i,t} = & \beta_1(PreEventCostlyExternalFinance_i \times PostSection711_{i,t}) \\ & + \beta_2 PostSection711_{i,t} + \gamma \frac{1}{Assets_{i,t-1}} + y_i + z_t + d_i \times z_t + \varepsilon_{i,t}. \end{aligned} \quad (3)$$

- $Hedging_{i,t}$ is log derivatives notional, a derivatives-use indicator, or the hedging ratio.
- $PreEventCostlyExternalFinance_i$ is high leverage or low distance-to-default.
- y_i are insurer fixed effects.
- z_t are year fixed effects.
- $d_i \times z_t$ are domicile-state-by-year fixed effects.
- The key coefficient is β_1 .

Interpretation: The coefficient β_1 compares more financially constrained insurers with less constrained insurers before and after Section 711, net of insurer fixed effects, year effects, and domicile-state-by-year shocks.

$$\begin{aligned} & Hedging_{i,t} \\ & = \beta_1 \cdot (Pre-event Costly External Finance \times PostSection711)_{i,Sec711} \\ & \quad + \beta_2 \cdot PostSection711_{i,Sec711} + \gamma \cdot 1/Assets_{i,t-1} + y_i + z_t + d_i \times z_t + \epsilon_{i,t} \end{aligned} \quad (1)$$

Original-paper excerpt: baseline hedging specification.

4.2 Baseline sales regression

The product-market specification is run at the insurer-state level:

$$\begin{aligned} Premiums_{i,s,t} = & \beta_1(PreEventCostlyExternalFinance_i \times PostSection711_{i,t}) \\ & + \beta_2 PostSection711_{i,t} + \gamma \frac{1}{Assets_{i,t-1}} + l_{i,s} + z_t + \varepsilon_{i,s,t}. \end{aligned} \quad (4)$$

- $Premiums_{i,s,t}$ is either log life insurance premiums or log annuity premiums sold by insurer i in licensing state s in year t .
- $l_{i,s}$ are insurer-licensing-state fixed effects.
- The coefficient β_1 measures whether treated insurers sell more in the states where they operate after their domicile state adopts Section 711.

Interpretation: This specification asks whether cheaper hedging changes real product-market outcomes, not merely balance-sheet risk management.

$$\begin{aligned} Premiums_{i,s,t} \\ = & \beta_1 \cdot (Pre-event Costly External Finance \times PostSection711)_{i,Sec711} \\ & + \beta_2 \cdot PostSection711_{i,Sec711} + \gamma \cdot 1/Assets_{i,t-1} + l_{i,s} + z_t + \epsilon_{i,s,t} \end{aligned} \quad (2)$$

Original-paper excerpt: premium/sales specification.

4.3 Fixed effects and comparison groups

- Insurer fixed effects absorb time-invariant differences across firms.
- Year fixed effects absorb aggregate shocks affecting all insurers.
- Domicile-state-by-year fixed effects absorb shocks specific to the state of domicile, including local regulatory or economic trends.
- Insurer-licensing-state fixed effects absorb persistent differences in where each insurer operates.
- Lagged $1/Assets$ controls for size because smaller insurers may face different financing conditions and growth opportunities.

Interpretation: The design is a rich difference-in-differences specification. It does not simply compare adopters to nonadopters; it compares more and less constrained insurers around the adoption event while controlling for insurer, year, and state-year effects.

4.4 Identification assumption

The identifying assumption is that, absent Section 711, high-leverage insurers and comparison insurers would have followed similar trends in hedging, financial stability, and product-market outcomes, after conditioning on fixed effects and controls.

- The reform is staggered across states.
- It is tied to insurer domicile, while many outcomes are measured across licensing states.

- Redomiciliation is rare, reducing concerns that firms choose domicile states to exploit the reform.
- Event-study plots and robustness checks examine pre-trends.
- Matching estimators and alternative treatment definitions help address treated-control differences.

Interpretation: The paper’s credibility comes from the combination of staggered legal adoption, heterogeneous exposure by pre-event financial fragility, fixed effects, event-study evidence, and robustness checks.

4.5 Why high leverage is the main exposure measure

- High leverage proxies for costly external finance and financial fragility.
- Derivatives counterparties are more likely to worry about legal enforceability when the insurer has higher default risk.
- If Section 711 matters through contract enforceability, the response should be strongest for high-leverage insurers.
- The paper also uses low distance-to-default to show that results are not specific to the leverage measure.

Interpretation: The exposure variable is not arbitrary. It is chosen because the legal reform should reduce hedging costs precisely for insurers whose counterparties were most concerned about default and receivership risk.

5 Results by section and subsection

5.1 Section 4.1: Derivatives usage after Section 711

- Table 3 is the first core result. High-leverage insurers increase hedging after Section 711.
- In the preferred specifications with lagged $1/Assets$, the interaction coefficient is 1.627 for log derivatives, 0.068 for the derivatives-use indicator, and 0.253 for the hedging ratio.
- The log coefficient implies an increase in derivatives notional of approximately 409%.
- The extensive-margin coefficient means the probability of using derivatives rises by about 6.8 percentage points.
- The hedging-ratio coefficient means the fraction of regulatory liabilities hedged rises by about 25.3 percentage points.
- Using low distance-to-default instead of high leverage produces similar findings; the hedging-ratio coefficient is 0.320.

Interpretation: The evidence supports the paper’s central mechanism: strengthening derivatives enforceability increases hedging for financially constrained insurers.

5.2 Section 4.1 continued: Risk-category evidence

- Table 4 separates derivatives by risk category: credit and foreign exchange, equity, interest rate, and interest-rate swaps.
- This split is important because actuarial guideline changes could affect equity-linked variable annuity hedging.

- The paper finds increases in categories not mainly driven by the actuarial-accounting change, especially credit/foreign-exchange and interest-rate derivatives.
- This helps separate the effect of Section 711 from other regulatory changes affecting variable annuity accounting.

Interpretation: The risk-category evidence is a mechanism robustness test. It shows the hedging response is not just an artifact of accounting changes for equity-linked annuities.

5.3 Section 4.2: Reinsurance

- Table 5 tests whether derivatives substitute for or complement reinsurance.
- Affiliated reinsurance declines for high-leverage insurers after Section 711: the coefficients are -0.040 and -0.069 in the two affiliated-reinsurance specifications.
- Unaffiliated reinsurance does not show a comparable increase.
- This suggests that some insurers move away from affiliated reinsurance when derivatives hedging becomes easier.

Interpretation: Reinsurance is another risk-management tool. The results suggest that derivatives become more attractive after the legal reform and partly displace affiliated reinsurance arrangements.

5.4 Section 4.3: Negative shocks and exit

- Table 6 shows that treated insurers become less likely to suffer large negative capital-and-surplus shocks.
- For high-leverage insurers, the propensity of capital & surplus falling below 50% of the prior-year level declines by 1.2 percentage points.
- Using low distance-to-default, the analogous decline is 1.6 percentage points.
- The propensity of exit due to receivership or other events also declines: -1.8 percentage points for high-leverage insurers and -2.0 percentage points for low-distance-to-default insurers.

Interpretation: This directly supports the financial-stability channel. Hedging does not merely change reported derivatives positions; it reduces downside balance-sheet outcomes and exit risk.

5.5 Section 4.3 continued: Hedging-adjusted income volatility

- Table 7 examines changes in hedging-adjusted income volatility.
- The coefficient on high leverage $\times$ PostSection711 is negative for volatility and positive for the indicator capturing volatility improvement.
- The low-distance-to-default specifications produce similar results.
- These estimates connect the paper to evidence that derivatives can improve hedging effectiveness.

Interpretation: If derivatives are actually used for risk management, income volatility should decline after the reform for exposed insurers. Table 7 shows this pattern.

5.6 Section 4.4: Ratings and financial strength

- Table 8 examines consumer-facing and regulatory measures of financial strength.
- Treated insurers become less likely to have low A.M. Best ratings.
- They become less likely to lose a top A.M. Best rating.
- IRIS and RBC measures improve.
- Parent-company put-option prices fall, indicating a decline in market-implied downside risk.

Interpretation: This section is crucial because insurance customers care about financial strength. Better ratings and lower distress proxies help explain why treated firms can sell more policies.

5.7 Section 4.5.1: Policy sales

- Table 9 shows that treated insurers increase policy sales after Section 711.
- In the baseline high-leverage specification, log life insurance premiums rise by 0.186 and log annuities rise by 0.360.
- These are economically meaningful sales increases, especially for annuity products.
- The effect persists after interacting treatment with firm size and alternative financial-fragility measures.

Interpretation: Hedging affects real product-market outcomes. Customers buy more insurance and annuity products from insurers whose financial position improves after hedging becomes easier.

5.8 Section 4.5.1 continued: Event-study evidence

- Figure 8 plots life insurance premiums and annuities around Section 711 adoption for treated versus control insurers.
- The authors acknowledge some evidence of pre-reform trends in the year before adoption.
- They use pre-trend assessment tests and robustness methods to address possible parallel-trend concerns.
- The post-reform coefficients remain positive and economically meaningful.

Interpretation: Event studies make the timing transparent. The paper does not hide pre-trend concerns; it explicitly assesses whether they can explain the main treatment effects.

5.9 Section 4.5.2: Matched-sample analysis

- Table 10 addresses the concern that treated and control insurers differ before adoption.
- The authors match high-leverage treated insurers to control insurers based on pre-event $1/Assets$ and net income, and then on $1/Assets$, net income, and leverage.
- The matched-sample results remain positive and significant.
- In the first matched design, life insurance premiums rise by 0.320 and annuities rise by 0.309.
- In the second matched design, life insurance premiums rise by 0.458 and annuities rise by 0.577.

Interpretation: The premium results are not driven solely by treated insurers being different kinds of firms. They survive a design that makes treated and control firms more comparable.

5.10 Section 4.5.4: Quantities and prices

- Table 11 examines whether higher premiums reflect more policies, higher prices, or both.
- The number of policies rises: the coefficient on high leverage $\times$ PostSection711 is 0.135.
- Prices fall: the coefficient is -0.034 for 10-year term life, -0.039 for life annuity, and -0.034 for term annuity.
- This pattern is consistent with the paper's conceptual diagram: supply shifts outward and competition increases, so quantities rise and prices fall.

Interpretation: The policy-sales increase is not just a price effect. It reflects a quantity expansion paired with lower prices, exactly what the competition mechanism predicts.

5.11 Section 4.6.1: Market concentration and product categories

- Table 12 splits outcomes by product type and market concentration.
- The effect is positive for ordinary life, group life, ordinary annuity, and group annuity products.
- For ordinary life and annuities, effects are stronger in more concentrated markets.
- This matters because the conceptual framework predicts larger product-market gains when the market is less perfectly competitive.

Interpretation: The stronger effect in concentrated markets supports the idea that risk management changes competition, not just firm-level sales mechanically.

5.12 Section 4.6.2: Market share

- Table 13 shows that treated insurers gain market share after Section 711.
- Life insurance market share rises by 0.056 percentage points and annuity market share rises by 0.235 percentage points.
- Treated insurers also become more likely to hold top market-share positions: the top-share indicator rises by 0.013 for life insurance and 0.072 for annuities.
- The industry is fragmented, so modest absolute market-share increases can still be meaningful.

Interpretation: The reform changes market structure at the margin. Financially constrained insurers become stronger competitors and capture more business.

5.13 Section 4.7: Performance

- Table 14 shows that operating income and net income increase for high-leverage insurers after Section 711.
- The operating-income-to-assets coefficient is 0.010.
- The net-income-to-assets coefficient is 0.011.
- This suggests that the sales expansion and price declines are not value destroying for treated insurers.

Interpretation: Hedging has both consumer and firm benefits in this setting. Consumers face lower prices and safer insurers; insurers gain sales, market share, and profitability.

6 Tables and figures

This section preserves the paper's original visual material. Adjacent figures and tables are grouped to save space and make comparisons faster.

6.1 Figures 1–3: Mechanism, derivatives usage, and reform timing

Read:

- Figure 1 gives the central economic intuition: lower financial risk shifts demand outward and lower financial input costs shift supply outward.
- Figure 2 shows the growth of life insurers' derivatives notional during the sample period.
- Figure 3 maps the staggered state adoption of Section 711, which supplies the paper's legal variation.

Economic message: The paper combines a conceptual mechanism with a staggered legal reform. The reform changes derivative-contract enforceability, and the mechanism predicts higher hedging, lower prices, and higher quantities for constrained insurers.

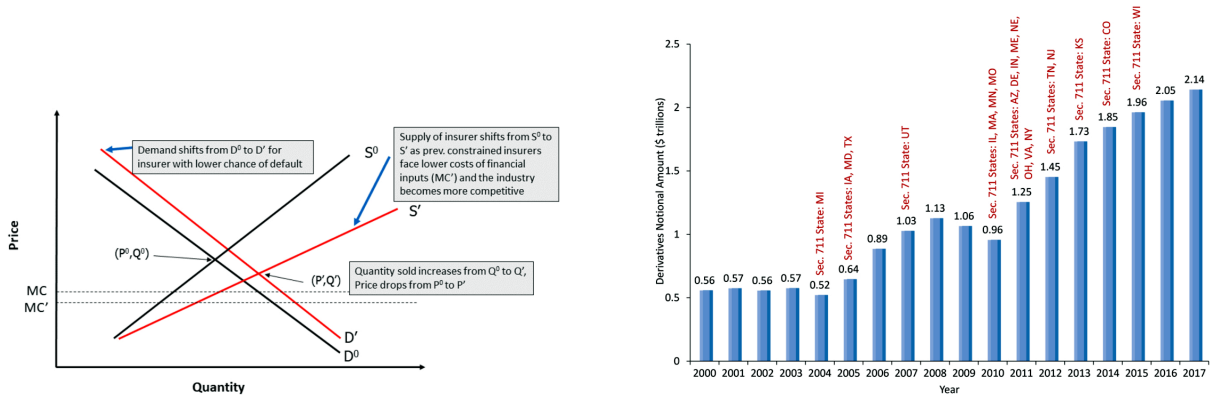
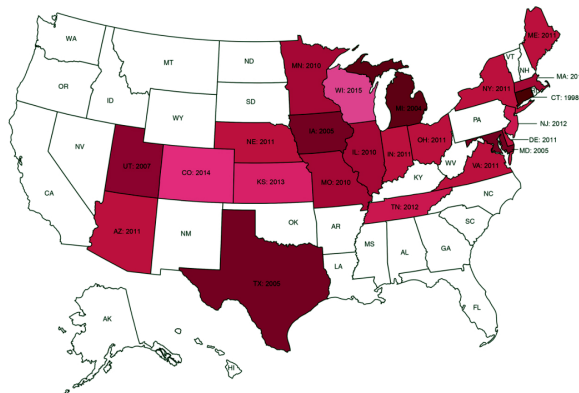


Figure 1: conceptual price-quantity effect of hedging.

Figure 2: derivatives notional amount of life insurers.



6.2 Figures 4–6 and Tables 1–2: Sample, geography, and variables

Read:

- Figure 4 shows where insurers are domiciled and licensed, emphasizing the difference between domicile states and product-market states.
- Figure 5 shows that many insurers operate in multiple states.
- Figure 6 shows that redomiciliation is rare after Section 711, helping support the design.
- Table 1 defines the core variables used throughout the paper.
- Table 2 reports summary statistics and shows that treated high-leverage insurers are larger, more derivative-intensive, and more levered than the broad sample.

Economic message: The legal treatment occurs through domicile states, while competitive effects are measured through the states in which firms sell policies. The geography and sample summaries are therefore central to the empirical design.

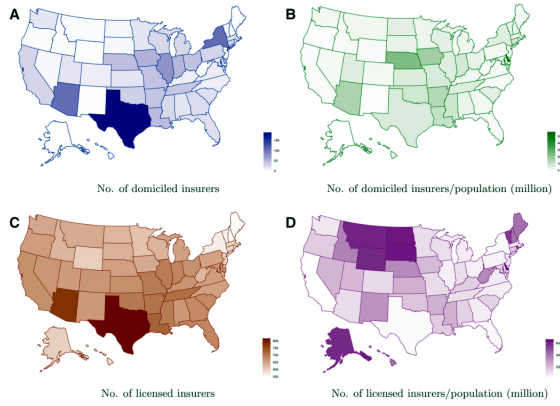


Figure 4: domicile and licensing-state heat maps.

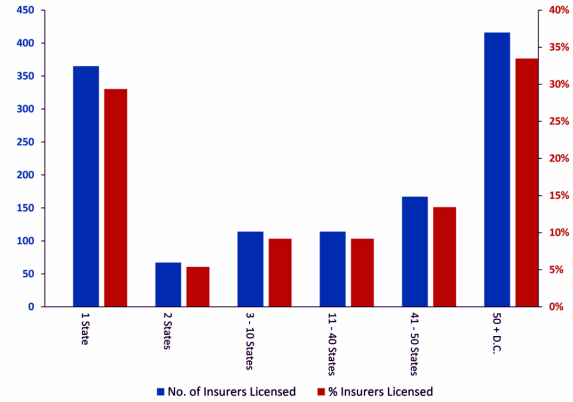


Figure 5: number of insurers licensed in one, two, or multiple states.

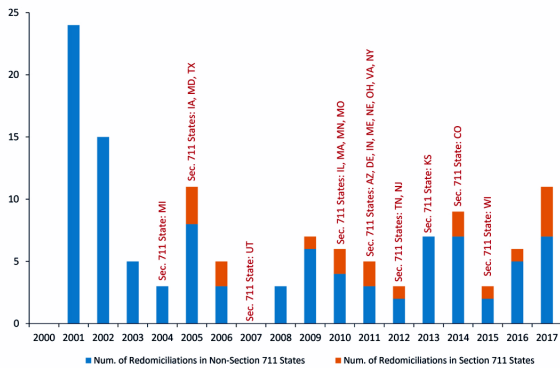


Figure 6: redomiciliations after Section 711.

Table 1 Key variables	
Variable	Definition
PostSection711	An indicator equals to one in the year of the passage of Section 711 by the insurer's domicile state and the following years, and zero otherwise. The variable is always zero for insurers that did not pass Section 711 during our sample period
Life insurance premiums	Life insurance premiums (SNL key field 121229)
Annuities	Total annuities related to mortality and morbidity risk (SNL key field 121230), annuities not incorporating mortality and morbidity risk (SNL key field 121231), and unallocated annuities (SNL key field 121232)
Derivatives notional	The notional amount of all derivatives contracts from the National Association of Insurance Commissioners (NAIC) Schedule DB
Derivatives (yes = 1)	An indicator for insurers reporting a derivatives notional value
Hedging ratio	The ratio of derivatives notional to total assets (SNL key field 122915) minus capital & surplus (SNL key field 122923)
Assets	Total assets (SNL key field 122915)
Net income	The ratio of net income (SNL key field 122937) to total assets
Leverage	The ratio of net liabilities to total assets, where net liabilities are calculated as total liabilities (SNL key field 122921) minus ceded reserves (SNL key fields' 121453 + 21451) plus assumed reserves (SNL key fields' 121439 + 121441)
Pre-event high leverage indicator	An indicator for insurers with Leverage above the sample median in the year before the insurer's domicile state adopted the Section 711
Distance-to-default	The ratio of operating cash flow (SNL key field 123445) minus liabilities (SNL key field 122921) to volatility of operating cash flows using 4 years of past consecutive observations
Pre-event low distance-to-default	An indicator for insurers with distance-to-default below the sample median in the year before the insurer's domicile state adopted the Section 711

Table 1: key variables.

Table 2
Summary statistics

	Mean	Median	SD	p25	p75	Obs.
<i>A. Insurer-state-level obs.</i>						
Life insurance premiums (\$millions)	5.967	0.177	30.443	0.007	1.949	383,382
Annuities (\$millions)	20.413	0.002	234.253	0.000	1.332	350,530
Life insurance premiums: Ordinary (\$millions)	4.775	0.107	24.305	0.004	1.357	367,663
Life insurance premiums: Credit (\$millions)	0.046	0.000	0.493	0.000	0.000	367,663
Life insurance premiums: Group (\$millions)	1.354	0.000	13.542	0.000	0.020	367,663
Life insurance premiums: Industrial (\$millions)	0.002	0.000	0.068	0.000	0.000	367,663
Annuities: Ordinary (\$millions)	10.292	0.001	57.754	0.000	0.870	333,678
Annuities: Credit (\$millions)	0.000	0.000	0.025	0.000	0.000	333,678
Annuities: Group (\$millions)	11.086	0.000	227.410	0.000	0.000	333,678
Annuities: Industrial (\$millions)	0.001	0.000	0.285	0.000	0.000	333,678
log of number of policies	6.006	6.217	2.866	3.932	8.132	334,794
<i>B. Insurer-level obs.</i>						
Life insurance premiums (\$millions)	208.888	10.418	818.688	0.978	83.792	12,047
Annuities (\$millions)	605.730	0.300	2,757.258	0.000	37.322	12,047
Derivatives notional (\$billions)	1.729	0.000	11.422	0.000	0.000	12,069
Derivatives (yes=1)	0.201	0.000	0.401	0.000	0.000	12,069
Hedging ratio	0.032	0.000	0.108	0.000	0.000	12,063
Assets (\$billions)	7.486	0.293	27.438	0.035	2.437	12,068
Leverage	0.679	0.748	0.637	0.460	0.884	12,045
Net income	0.014	0.008	0.113	0.000	0.025	12,068
log of 10-year term life policy price	6.268	6.160	0.723	5.635	6.858	10,155
log of life annuity policy price	9.708	9.705	0.172	9.590	9.827	6,621
log of term annuity policy price	9.338	9.448	0.433	9.113	9.655	1,415
<i>C. Treated group: Pre-event high leverage insurers</i>						
<i>C1. Insurer-state-level obs.</i>						
Life insurance premiums (\$millions)	9.689	0.400	42.449	0.024	3.835	177,155
Annuities (\$millions)	34.391	0.139	319.475	0.000	7.923	163,945
Life insurance premiums: Ordinary (\$millions)	7.698	0.256	33.358	0.015	2.905	173,923
Life insurance premiums: Credit (\$millions)	0.044	0.000	0.506	0.000	0.000	173,923
Life insurance premiums: Group (\$millions)	2.062	0.000	18.386	0.000	0.056	173,923
Life insurance premiums: Industrial (\$millions)	0.002	0.000	0.055	0.000	0.000	173,923
Annuities: Ordinary (\$millions)	17.409	0.090	76.915	0.000	4.325	159,589
Annuities: Credit (\$millions)	0.000	0.000	0.001	0.000	0.000	159,589
Annuities: Group (\$millions)	17.810	0.000	307.045	0.000	0.092	159,589
Annuities: Industrial (\$millions)	0.002	0.000	0.412	0.000	0.000	159,589
log of number of policies	6.405	6.592	2.806	4.431	8.460	163,136
<i>C2. Insurer-level obs.</i>						
Derivatives notional (\$billions)	3.937	0.000	17.784	0.000	0.138	4,409
Derivatives (yes=1)	0.347	0.000	0.476	0.000	1.000	4,409
Hedging ratio	0.059	0.000	0.142	0.000	0.022	4,409
Assets (\$billions)	15.989	1.669	41.758	0.303	10.040	4,409
Leverage	0.855	0.860	0.227	0.778	0.926	4,406
Net income	0.006	0.006	0.036	0.001	0.014	4,409
log of 10-year term life policy price	6.275	6.179	0.717	5.638	6.855	6,259
log of life annuity policy price	9.723	9.727	0.164	9.606	9.842	4,839
log of term annuity policy price	9.364	9.463	0.431	9.129	9.676	1,024

(Continued)

Table 2
(Continued)

	Mean	Median	SD	p25	p75	Obs.
<i>D. Control group: Pre-event low leverage insurers & High/low leverage insurers in non-Section-711 states</i>						
<i>D1. Insurer-state-level obs.</i>						
Life insurance premiums (\$millions)	2.770	0.085	12.362	0.002	0.989	206,227
Annuities (\$millions)	8.131	0.000	114.403	0.000	0.053	186,585
Life insurance premiums: Ordinary (\$millions)	2.150	0.044	10.389	0.001	0.616	193,740
Life insurance premiums: Credit (\$millions)	0.049	0.000	0.481	0.000	0.000	193,740
Life insurance premiums: Group (\$millions)	0.718	0.000	6.612	0.000	0.006	193,740
Life insurance premiums: Industrial (\$millions)	0.003	0.000	0.078	0.000	0.000	193,740
Annuities: Ordinary (\$millions)	3.768	0.000	29.685	0.000	0.039	174,089
Annuities: Credit (\$millions)	0.000	0.000	0.034	0.000	0.000	174,089
Annuities: Group (\$millions)	4.922	0.000	112.339	0.000	0.000	174,089
Annuities: Industrial (\$millions)	0.000	0.000	0.000	0.000	0.000	174,089
log of number of policies	5.628	5.864	2.871	3.434	7.799	171,658
<i>D2. Insurer-level obs.</i>						
Derivatives notional (\$billions)	0.457	0.000	4.371	0.000	0.000	7,660
Derivatives (Yes=1)	0.116	0.000	0.321	0.000	0.000	7,660
Hedging ratio	0.016	0.000	0.078	0.000	0.000	7,660
Assets (\$billions)	2.591	0.090	10.817	0.019	0.625	7,659
Leverage	0.577	0.582	0.762	0.345	0.817	7,639
Net Income	0.018	0.012	0.139	-0.001	0.036	7,659
log of 10-year term life policy price	6.257	6.144	0.732	5.611	6.858	3,896
log of life annuity policy price	9.669	9.663	0.186	9.556	9.770	1,782
log of term annuity policy price	9.272	9.383	0.432	9.064	9.615	391

Table 2, page 1: summary statistics.

Table 2, page 2: summary statistics continued.

6.3 Tables 3–4 and Figure 7: Hedging response

Read:

- Table 3 shows that high-leverage insurers increase derivatives notional, the probability of using derivatives, and the hedging ratio after Section 711.
- Table 4 shows that the response is present across economically relevant derivatives categories, not only in equity-linked annuity hedges.
- Figure 7 shows event-time patterns for hedging outcomes around Section 711 adoption.

Economic message: These are the key first-stage results: the legal reform reduces the effective cost of derivatives and causes constrained insurers to hedge more.

Table 3 Derivatives usage of life insurance companies after Section 711 adoption								
Dep. variable	log of Derivatives		Derivatives (yes = 1)			Hedging ratio		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pre-event high leverage × PostSection711	1.612** (0.653)	1.627** (0.651)		0.060** (0.031)	0.068** (0.031)		0.233** (0.091)	0.253** (0.091)
Pre-event low distance-to-default × PostSection711			1.460** (0.658)		0.038* (0.032)			0.320** (0.090)
PostSection711	-1.217 (0.895)	-1.213 (0.897)	-1.202 (0.951)	-0.038 (0.039)	-0.036 (0.040)	-0.173 (0.184)	-0.226 (0.183)	
Lagged 1/Assets			-0.142** (0.059)	-0.130** (0.064)	-0.010** (0.004)	-0.009** (0.004)	0.002 (0.009)	-0.004 (0.009)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Domicile state × Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-event high leverage	Absorbed	Absorbed	NA	Absorbed	Absorbed	NA	Absorbed	NA
Pre-event low distance-to-default	NA	NA	Absorbed	NA	Absorbed	NA	Absorbed	NA
Observations	11,861	11,774	11,774	11,847	11,860	11,841	11,857	11,857
Number of companies	809	809	809	806	800	806	800	800
Adjusted R ²	809	809	809	746	746	746	623	624

This table presents estimations from derivatives regressions. The sample includes life insurance company-level data for the period 2000-2017. The dependent variable in columns 1-3 is $\log$ of derivatives, which is defined as the natural logarithm of the yearly derivatives notional value. The dependent variable in columns 4-5 is $\text{Derivatives (yes = 1)}$, which is an indicator for insurers reporting a derivatives notional value. The dependent variable in columns 6-8 is Hedging ratio , which is the ratio of derivatives notional to total assets minus capital & surplus. Leverage is net liabilities/total assets and $\text{Distance-to-default}$ is operating cash flows - liabilities/volatility of past operating cash flows. Both are measured in the year prior to Section 711 passage. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity robust and double-clustered at the domicile-state and year levels, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 4 Derivatives usage of life insurance companies after Section 711 adoption: Different risk categories							
Dep. variable	CR + FX		EQ		IR		B. Hedging exposure Regulatory capital
	(1)	(2)	(3)	(4)	(5)	(6)	IR Swaps (8)
Pre-event high leverage × PostSection711	1.222*** (0.326)		1.650** (0.645)		1.166*** (0.370)		0.030** (0.007)
Pre-event low distance-to-default × PostSection711			0.759** (0.347)		1.628** (0.631)		1.031* (0.491)
PostSection711	-1.835** (0.807)	-1.820** (0.790)	-1.620* (0.821)	-1.669* (0.844)	-1.698* (0.844)	-1.667* (0.917)	-0.014 (0.015)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Domicile state × Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-event high leverage	Absorbed	NA	Absorbed	NA	Absorbed	NA	Absorbed
Pre-event low distance-to-default	NA	Absorbed	NA	Absorbed	NA	Absorbed	NA
Observations	11,819	11,819	11,798	11,798	11,815	11,815	11,860
Number of companies	800	800	800	800	800	800	800
Adjusted R ²	729	728	720	720	779	779	408

This table presents estimations from derivatives regressions, with derivatives categorized by risk type. The sample includes life insurance company-level data for the period 2000-2017. The dependent variable in columns 1 and 2 is the natural logarithm of the yearly notional value of the sum of credit risk (CR) and foreign exchange (FX) derivatives. The dependent variables in columns 3-4 and 5-6 are the natural logarithm of the yearly notional value of equity (EQ) derivatives and interest rate (IR) derivatives, respectively. The dependent variable in columns 7 and 8 is the ratio of hedging exposure computed for IR swaps scaled by capital & surplus. All regressions control for lagged 1/Assets. Leverage is net liabilities/total assets and $\text{Distance-to-default}$ is operating cash flows - liabilities/volatility of past operating cash flows. Both are measured in the year prior to Section 711 passage. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity robust and double-clustered at the domicile-state and year levels, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 3: derivatives usage after Section 711.

Table 4: derivatives usage by risk category.

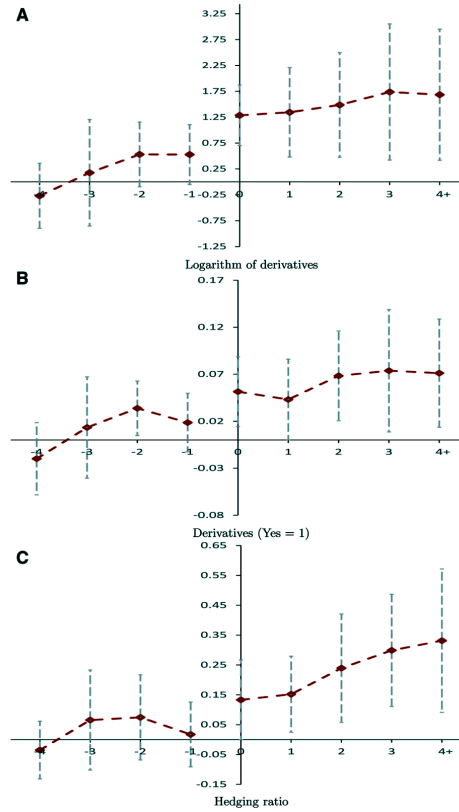


Figure 7: event-study evidence for derivatives usage.

6.4 Tables 5–8: Risk management, risk reduction, and financial strength

Read:

- Table 5 shows that affiliated reinsurance declines after Section 711 for treated high-leverage insurers.
- Table 6 shows that negative capital shocks and exits decline.
- Table 7 shows that hedging-adjusted income volatility improves.
- Table 8 shows that ratings and financial-strength measures improve.

Economic message: The reform does not merely raise derivative use. It improves financial stability, which is necessary for the product-market channel.

Table 5
Reinsurance of life insurance companies after Section 711 adoption

Dep. variables:	Affiliated reinsurance		Unaffiliated reinsurance	
	Reinsurance/ Assets (1)	Reinsurance/ (Assets - Capital & surplus) (2)	Reinsurance/ Assets (3)	Reinsurance/ (Assets - Capital & surplus) (4)
Pre-event high leverage $\times$ PostSection711	-0.040** (0.017)	-0.069** (0.024)	0.004 (0.017)	-0.001 (0.026)
PostSection711	-0.009 (0.038)	0.022 (0.046)	-0.021 (0.026)	-0.012 (0.036)
Lagged 1/Assets	0.490 (0.038)	1.056 (0.666)	-0.006 (0.046)	0.086 (0.108)
Year fixed effects	Yes	Yes	Yes	Yes
Domicile state $\times$ Year fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes
Pre-event high leverage	Absorbed	Absorbed	Absorbed	Absorbed
Observations	4,383	4,381	8,283	8,281
Number of companies	374	374	733	733
Adjusted R^2	.610	.496	.665	.631

This table presents estimations from reinsurance regressions. The sample includes life insurance companies with available reinsurance-agreement-level data for the period 2004–2017. The dependent variables in columns 1, 2, 3, and 4 are the ratio of affiliated reinsurance to assets, the ratio of affiliated reinsurance to assets minus capital & surplus, the ratio of unaffiliated reinsurance to assets, and the ratio of unaffiliated reinsurance to assets minus capital & surplus, respectively. Reinsurance is defined as the reinsurance ceded by the life insurance companies and measured as the sum of reserve credit taken on ceded life insurance (SNL key field 120744) plus modified coinsurance reserve ceded (SNL key field 120742). For each reinsurance agreement, we observe whether the reinsurance relationship is with an affiliated entity or not (SNL key field 124313). We use this information to identify affiliated and unaffiliated reinsurance. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity-robust and double-clustered at the domicile-state and year levels, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 5: reinsurance after Section 711.

Table 7
Change in hedging adjusted income volatility after Section 711 adoption

Dep. variables:	Δ Volatility		Δ Volatility dummy	
	(1)	(2)	(3)	(4)
Pre-event high leverage $\times$ PostSection711	-0.0004** (0.0002)		0.066** (0.023)	
Pre-event low distance-to-default $\times$ PostSection711		-0.0004* (0.0002)		0.078** (0.032)
PostSection711	0.0004 (0.0003)	0.0004 (0.0003)	-0.046 (0.045)	-0.057 (0.049)
Year fixed effects	Yes	Yes	Yes	Yes
Domicile state $\times$ Year fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes
Pre-event high leverage	Absorbed	NA	Absorbed	NA
Pre-event low distance-to-default	NA	Absorbed	NA	Absorbed
Observations	11,797	11,797	11,860	11,860
Number of companies	878	878	880	880
Adjusted R^2	.335	.335	.475	.476

This table presents estimations from change in hedging adjusted income volatility regressions. The sample includes life insurance company-level data for the period 2000–2017. The dependent variable in columns 1 and 2, Δ Volatility, is defined as volatility of net income minus volatility of adjusted net income. Volatility of (adjusted) net income is the standard deviation of (adjusted) net income using 4 years of past consecutive observations to the average value of assets estimated over the same period. Adjusted net income is defined as net income minus the sum of net investment income and total capital gains from derivatives (Koijen and Yogo 2017). The dependent variable in columns 3 and 4, Δ Volatility dummy, is an indicator that takes the value 1 if Δ Volatility < 0 . All regressions control for lagged 1/Assets. *Leverage* is net liabilities/total assets and *Distance-to-Default* is (operating cash flows – liabilities)/volatility of past operating cash flows. Both are measured in the year prior to Section 711 passage. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity-robust and double-clustered at the domicile-state and year levels, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 7: hedging-adjusted income volatility.

Table 6
Propensity of shocks to capital & surplus and exit after Section 711 adoption

Dep. variables:	A. Propensity of negative capital & surplus shocks		B. Propensity of exit	
	Capital & surplus $< 50\%$ of previous year capital & surplus dummy (yes = 1)		Exit due to receivership or other events dummy (yes = 1)	
	(1)	(2)	(3)	(4)
Pre-event high leverage $\times$ PostSection711	-0.012*** (0.004)		-0.018** (0.008)	
Pre-event low distance-to-default $\times$ PostSection711		-0.016** (0.006)		-0.020*** (0.005)
PostSection711	0.001 (0.006)	0.004 (0.006)	-0.010 (0.026)	-0.008 (0.026)
Year fixed effects	Yes	Yes	Yes	Yes
Domicile state $\times$ Year fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes
Pre-event high leverage	Absorbed	NA	Absorbed	NA
Pre-event low distance-to-default	NA	Absorbed	NA	Absorbed
Observations	11,827	11,827	11,860	11,860
Number of companies	878	878	880	880
Adjusted R^2	.066	.067	.130	.130

This table presents estimations from negative capital & surplus shock regressions (panel A) and exit regression (panel B). The sample includes life insurance company level data for the period 2000–2017. The dependent variable in columns 1 and 2 of panel A is an indicator for insurers with Capital & Surplus (the ratio of capital and surplus to total assets) in the current period less than 50% of Capital & Surplus in the previous period. The dependent variable in columns 3 and 4 of panel B is an indicator for insurers placed in receivership or exiting the sample. All regressions control for lagged 1/Assets. *Leverage* is net liabilities/total assets and *Distance-to-default* is (operating cash flows – liabilities)/volatility of past operating cash flows. Both are measured in the year prior to Section 711 passage. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity-robust and double-clustered at the domicile-state and year levels, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 6: capital shocks and exit.

Table 8
Financial strength after Section 711 adoption

Dep. variables	Low A.M. Best rating (1)	Losing top A.M. Best rating (2)	Low IRIS metric (3)	Low RBC ratio (4)	Put option price (5)
Pre-event high leverage $\times$ PostSection711	-0.005** (0.002)	-0.023*** (0.007)	-0.077*** (0.026)	-0.059* (0.034)	-2.135* (1.192)
PostSection711	-0.012 (0.008)	0.021** (0.009)	-0.053 (0.043)	0.014 (0.046)	1.275 (0.995)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Domicile state $\times$ Year fixed effects	Yes	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes	Yes
Pre-event high leverage	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed
Observations	6,655	3,661	11,860	11,657	1,216
Number of companies	471	352	880	871	145
Adjusted R^2	.587	.025	.208	.491	.227

This table presents estimations from financial strength regressions. The samples in columns 1–4 and column 5 include life insurance company-level data and parent company data, respectively, for the period 2000–2017. The dependent variable in column 1 is an indicator for firms with an A.M. Best rating equal to or less than C+-. The dependent variable in column 2 is a dummy equal to one for firms with a rating in one of the B categories (B+, B+, B, and B-) in period t , but with a top A++ or A+ rating at $t-1$. In this test, we focus exclusively on firms with A++ and A+ ratings (the top 2 A.M. Best ratings) and B+, B+, B, and B- ratings. The dependent variable in column 3 is a dummy equal to one if the number of IRIS ratios below the adequacy level is above the sample median. The dependent variable in column 4 is an indicator for insurers with an RBC ratio below the sample median. The dependent variable in column 5 is the ratio of put option price to stock price minus strike price. All regressions control for lagged 1/Assets. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity-robust and double-clustered at the domicile-state and year levels, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 8: financial strength.

6.5 Table 9, Figure 8, and Table 10: Policy sales and matched samples

Read:

- Table 9 shows higher life insurance premiums and annuity premiums for treated insurers after Section 711.
- Figure 8 plots event-time premium and annuity effects, making the timing transparent.
- Table 10 uses matched samples and obtains similar positive premium effects.

Economic message: The paper’s central real-outcome result is that improved hedging access translates into higher sales. The matched-sample evidence helps rule out simple treated-control composition explanations.

Dep. variables	log of life insurance premiums (1)	log of annuities (2)	log of life insurance premiums (3)	log of annuities (4)	log of life insurance premiums (5)	log of annuities (6)
Pre-event high leverage × PostSection711	0.186*** (0.035)	0.369*** (0.067)				
Pre-event large firm × Pre-event high leverage × PostSection711			0.172*** (0.033)	0.396*** (0.072)		
Pre-event small firm × Pre-event high leverage × PostSection711			0.268** (0.097)	0.511*** (0.102)		
Pre-event high concentration × Pre-event high leverage × PostSection711					0.215*** (0.047)	0.426*** (0.084)
Pre-event low concentration × Pre-event high leverage × PostSection711					0.186*** (0.046)	0.303*** (0.055)
PostSection711	-0.041 (0.040)	-0.325*** (0.047)	-0.037 (0.040)	-0.354*** (0.052)	-0.040 (0.040)	-0.326*** (0.046)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Insurer licensing-state fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Insurer fixed effects	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed
Pre-event high leverage	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed
Observations	373,425	339,205	373,425	339,205	363,410	337,597
Number of companies	849	849	849	849	849	849
Number of company-state obs.	28,033	25,717	28,033	25,717	27,934	25,686
Adjusted R ²	.926	.902	.926	.902	.926	.902

This table presents estimations from life insurance and annuity premium regressions. The sample includes life insurance company-state-level data for the period 2000–2017. The dependent variable in columns 1, 3, and 5 is *log of life insurance premiums*, which is defined as the natural logarithm of life insurance premiums. The dependent variable in columns 2, 4, and 6 is the *log of annuities*, which is defined as the natural logarithm of total annuities. Pre-event Large Firm is an indicator for insurers with *1/Assets* below the sample median in the year before the insurer's domicile state adopted the Section 711. Pre-event Small Firm is an indicator for insurers with *1/Assets* above the sample median in the year before the insurer's domicile state adopted the Section 711. Pre-event High Concentration is an indicator for insurers with mean *HHI* across licensing states above the sample 75th percentile in the year before the insurer's domicile state adopted the Section 711. Pre-event Low Concentration is an indicator for insurers with mean *HHI* across licensing states below the sample 75th percentile in the year before the insurer's domicile state adopted the Section 711. Pre-event Large Firm, Pre-event Small Firm, Pre-event High Concentration, and Pre-event Low Concentration are always zero for firms in the control states. All regressions control for lagged *1/Assets*. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity-robust and double-clustered at the licensing-state and year levels, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 9: life insurance premiums and annuities after Section 711.

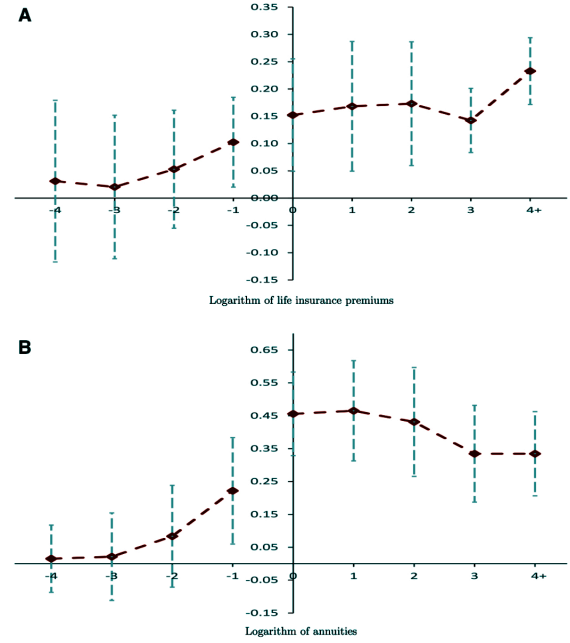


Figure 8: event-study evidence for premiums and annuities.

Dep. variables	A. Matching based on pre-event <i>1/Assets</i> and net income		B. Matching based on pre-event <i>1/Assets</i> , net income, and leverage	
	log of life insurance premiums (1)	log of annuities (2)	log of life insurance premiums (3)	log of annuities (4)
Pre-event high leverage × PostSection711	0.320*** (0.051)	0.309*** (0.087)	0.458*** (0.089)	0.577*** (0.082)
PostSection711	-0.246*** (0.061)	-0.380*** (0.070)	-0.382*** (0.115)	-0.601*** (0.080)
Year fixed effects	Yes	Yes	Yes	Yes
Insurer licensing-state fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	Absorbed	Absorbed	Absorbed	Absorbed
Pre-event high leverage	Absorbed	Absorbed	Absorbed	Absorbed
Observations	248,013	188,654	226,011	170,054
Number of companies	375	373	343	340
Number of company-state obs.	15,548	12,791	14,138	11,445
Adjusted R ²	.935	.899	.930	.895

This table presents estimations from life insurance premium and annuity regressions. In any given Section 711 event year, we match each pre-event high leverage insurer (treated) to its closest control insurer based on pre-event *1/Assets* and net income (panel A) and *1/Assets*, net income, and leverage (panel B) using the [Abadie and Imbens' \(2006\)](#) bias-corrected matching estimator. All regressions control for lagged *1/Assets*. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity-robust and double-clustered at the licensing-state and year levels, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 10: matched-sample premium regressions.

6.6 Tables 11–14: Prices, competition, market share, and profitability

Read:

- Table 11 shows that treated insurers sell more policies and lower prices for term life and annuity products.
- Table 12 shows stronger effects in more concentrated markets and across ordinary/group product categories.
- Table 13 shows that treated insurers gain market share and become more likely to hold top market-share positions.
- Table 14 shows that operating income and net income rise.

Economic message: The product-market mechanism is complete: treated insurers hedge more, become safer, sell more, lower prices, gain share, and earn higher income.

Table 11
Quantity and pricing of life insurance products after Section 711 adoption

Dep. variable:	log of number of policies	log of policy prices		
	Number of policies (1)	10-year term life policy (2)	Life annuity (3)	Term annuity (4)
Pre-event high leverage × PostSection711	0.135*** (0.034)	−0.034* (0.017)	−0.039*** (0.011)	−0.034*** (0.005)
PostSection711	−0.047 (0.035)	0.029 (0.038)	−0.010 (0.026)	0.021 (0.037)
Product fixed effects	NA	Yes	Yes	Yes
Gender fixed effects	NA	Yes	Yes	NA
Year fixed effects	Yes	Yes	Yes	Yes
Insurer-licensing-state fixed effects	Yes	NA	NA	NA
Domicile state × Year fixed effects	No	Yes	Yes	Yes
Insurer fixed effects	Absorbed	Yes	Yes	Yes
Pre-event high leverage	Absorbed	Absorbed	Absorbed	Absorbed
Observations	334,794	10,155	6,621	1,415
Number of companies	785	154	53	46
Number of company-state obs.	25,759	NA	NA	NA
Adjusted R ²	.936	.971	.873	.989

This table presents estimations examining the number of policies (column 1) and prices for different insurance products including 10-year term life policies (column 2), life annuity (column 3) and term annuities (columns 4). All regressions control for lagged 1/Assets. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity-robust and double-clustered at the licensing-state and year and domicile-state and year levels in column 1 and columns 2–4, respectively, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 11: product quantities and prices.

Table 13
Market share after Section 711 adoption

Dep. variables:	A. Market share		B. Top market share	
	Market share: Life insu. premiums (1)	Market Share: Annuities (2)	Life insu. prem. > 75 th %tile dummy (yes = 1) (3)	Annuity > 75 th %tile dummy (yes = 1) (4)
Pre-event high leverage × PostSection711	0.056*** (0.010)	0.235*** (0.039)	0.013** (0.005)	0.072*** (0.017)
PostSection711	−0.008 (0.006)	−0.074*** (0.021)	0.006 (0.005)	−0.066*** (0.014)
Year fixed effects	Yes	Yes	Yes	Yes
Insurer-licensing-state fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	Absorbed	Absorbed	Absorbed	Absorbed
Pre-event high leverage	Absorbed	Absorbed	Absorbed	Absorbed
Observations	338,680	196,433	338,680	196,433
Number of companies	790	578	790	578
Number of company-state obs.	25,948	16,869	25,948	16,869
Adjusted R ²	.803	.637	.818	.756

This table presents estimations from market share and top market share regressions. The sample includes life insurance company-state-level data for the period 2000–2017. The dependent variable in column 1 is the ratio of life insurance premiums to total life insurance premiums collected by all the insurers in each state-year. The dependent variable in column 2 is the ratio of annuities to total annuities collected by all the insurers in each state-year. We multiply the dependent variables in columns 1 and 2 by 100. The dependent variables in columns 3 and 4 are indicators for life insurers with life insurance premiums and annuities, respectively, above the respective state-year sample 75th percentile. All regressions control for lagged 1/Assets. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity-robust and double-clustered at the licensing-state and year levels, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 13: market share after Section 711.

Table 12
Life insurance premiums and annuities after Section 711 adoption: Different categories

Dep. variables:	A. Life insurance premiums		B. Annuities	
	log of ordinary life (1)	log of group life (2)	log of ordinary annuity (3)	log of group annuity (4)
Pre-event high concentration × Pre-event high leverage	0.258*** (0.046)	0.181** (0.068)	0.475*** (0.089)	0.244*** (0.047)
× PostSection711	0.181*** (0.039)	0.171*** (0.036)	0.330*** (0.078)	0.085 (0.053)
Pre-event low concentration × Pre-event high leverage	−0.075* (0.037)	−0.071** (0.030)	−0.269*** (0.040)	−0.114** (0.041)
PostSection711	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Insurer-licensing-state fixed effects	Absorbed	Absorbed	Absorbed	Absorbed
Insurer fixed effects	Absorbed	Absorbed	Absorbed	Absorbed
Pre-event high leverage	359,722	359,722	331,876	331,876
Observations	833	833	836	836
Number of companies	27,542	27,542	25,276	25,276
Number of company-state obs.	.939	.879	.889	.876
Adjusted R ²				

This table presents estimations from different life insurance (panel A) and annuity (panel B) regressions. The sample includes life insurance company-state-level data for the period 2000–2017. The dependent variables in columns 1 and 2 are the natural logarithm of ordinary life insurance premiums (SNL key field 121302), respectively. The dependent variables in columns 3 and 4 are the natural logarithm of ordinary annuities (SNL key fields 121320 + 121325 + 121328) and group annuities (SNL key fields 121322 + 121326 + 121330), respectively. Pre-event High Concentration is an indicator for insurers with mean HHI across licensing states above the sample 75th percentile in the year before the insurer's domicile state adopted the Section 711. Pre-event Low Concentration is an indicator for insurers with mean HHI across licensing states below the sample 75th percentile in the year before the insurer's domicile state adopted the section 711. Pre-event High concentration and Pre-event Low Concentration are always zero for firms in the control states. All regressions control for lagged 1/Assets. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity-robust and double-clustered at the licensing-state and year levels, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 12: product categories and concentration.

Table 14
Income of life insurance companies after Section 711 adoption

Dep. variables:	Operating income (1)	Net income (2)
Pre-event high leverage × PostSection711	0.010* (0.005)	0.011** (0.004)
PostSection711	0.002 (0.008)	−0.005 (0.007)
Year fixed effects	Yes	Yes
Domicile state × Year fixed effects	Yes	Yes
Insurer fixed effects	Yes	Yes
Pre-event high leverage	Absorbed	Absorbed
Observations	11,835	11,860
Number of companies	880	880
Adjusted R ²	.248	.111

This table presents estimations from income regressions. The sample includes life insurance company-level data for the period 2000–2017. The dependent variable in column 1 is the ratio of operating income to total assets. The dependent variable in column 2 is the ratio of net income to total assets. All regressions control for lagged 1/Assets. Refer to Table 1 for detailed variable definitions. Standard errors are heteroscedasticity-robust and double-clustered at the domicile-state and year levels, and reported inside parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 14: income after Section 711.

7 Short summary

- The paper studies whether cheaper hedging affects firm behavior and competition.
- The setting is U.S. life insurers from 2000 to 2017.
- The shock is staggered adoption of Section 711 of the Insurer Receivership Model Act.
- Section 711 gives derivatives counterparties superpriority in insurer insolvency.
- This improves derivatives contract enforceability and lowers counterparty concerns.
- The reform should matter most for high-leverage insurers and insurers closer to default.
- Treated insurers increase derivatives notional, derivatives adoption, and hedging ratios.
- The hedging response is robust to alternative financial-constraint measures and derivatives categories.
- Treated insurers reduce affiliated reinsurance, consistent with derivatives becoming more attractive.

- They experience fewer large negative capital-and-surplus shocks and lower exit risk.
- Their hedging-adjusted income volatility improves.
- Ratings and financial-strength proxies improve.
- They sell more life insurance and annuity products.
- They lower prices for several products and increase the number of policies sold.
- They gain market share, particularly in annuities and in more concentrated markets.
- Operating income and net income rise after the reform.
- The paper concludes that risk management affects competition by changing the financial stability and competitive behavior of constrained firms.

8 Conclusion

8.1 Core conclusion

Giambona, Kumar, and Phillips show that the enforceability of hedging contracts can have real competitive effects. A legal reform that protects derivatives counterparties in insurer insolvency raises hedging by financially constrained life insurers. This reduction in risk improves financial strength and allows treated firms to sell more policies, lower prices, and gain market share.

8.2 Economic intuition

Hedging is valuable for fragile insurers because it reduces the probability of financial distress. But fragile insurers may have difficulty accessing derivatives when counterparties worry about default and collateral recovery. Section 711 reduces those concerns. Once hedging becomes easier, constrained insurers become safer. Because insurance customers care about insurer solvency, safer insurers can attract demand. Because risk-bearing costs fall, they can also price more aggressively.

8.3 Identification takeaway

The empirical design uses staggered state adoption and differential exposure by pre-event financial fragility. The main coefficient compares high-leverage insurers to less exposed insurers around the legal change, while absorbing insurer, year, and state-year variation. Event studies, alternative exposure measures, matched samples, and robustness tests support the interpretation that Section 711 changes hedging and downstream product-market outcomes.

8.4 Broader takeaway

The paper links financial contracting to industrial organization. Risk management is not merely a treasury function; it can alter product-market competition. Legal rules governing derivatives priority affect who can hedge, how stable firms become, and how aggressively they compete.